

Report

Problem Description:

Existing weather applications usually only have one or two functions so the issue here is their limited functionality and that is that they display the temperature or description of the weather of a certain city or , that the user interfaces were too complicated for users to use which makes it hard for people to use weather applications , Recognizing these challenges, our team thought to recreate weather applications, and our aim and mission was to approach weather applications in a different way , and that is to make it more interesting , and to make people more intrigued with our application

1. `window()`: Yara mahmoud
2. `entertainment_menu()`: Yara mahmoud
3. `data_menu()`: zeina mohamed
4. `weather_encoding_decoding_game()`: zeina mohamed
5. `weather_quiz()`: zeina mohamed
6. `read_user_score(game_type)`: zeina mohamed
7. `save_user_score(game_type, new_score)`: zeina mohamed
8. `get_weather_data(city)`: Yara mahmoud
9. `save_data_to_file(city, option, columns, data)`: Yara mahmoud
10. `data_options(city)`: zeina mohamed
11. `weather_forecast_option(city)`: Yara mahmoud
12. `save_forecast_data_to_file(city, num_days, columns, grouped_forecast)`: Yara mahmoud
13. `compare_weather_data_option(initial_city)`: Yara mahmoud
14. `save_compare_data_to_file(initial_city, cities, headers, data)`: Yara mahmoud
15. `compare_table(data, headers)`: Yara mahmoud
16. `highlight_highest_temperature(tree)`: zeina mohamed
17. `temperature_table(city)`: Yara mahmoud
18. `wind_table(city)`: Yara mahmoud
19. `show_sun_time(city, time_type)`: zeina mohamed
20. `temperature_in_units(city)`: Yara mahmoud
21. `wind_speed_in_units(city)`: Yara mahmoud
22. `get_sun_time(city, time_type)`: zeina mohamed
23. `getnotification(temperature, weather_description)`: zeina mohamed

Our approach involved constant collaboration and shared ideation, ensuring that no team member faced isolation in tackling specific functions. as a result of the project functions being so linked and based on each other, a function being written by a member of the team does not mean that she was completely left alone to deal with the function, we collectively brainstormed innovative ideas and systematically transformed them into the cohesive code structure. Our collaboration extended beyond individual responsibilities, collaboration was achieved in this project as we were both thinking together about the ideas and the systematic ways to apply them and turn the ideas into the written code you just saw. We were both thinking and searching together about the algorithm to achieve the required code and to solve the problems we faced during testing the code.

How can this problem be solved?

The solution we created is to analyze the weather data and give the user a set of features and options to make his choice from, we also added entertainment part where the user can take a weather quiz that have couple of questions about the weather and also fun encoding and decoding game that displays shuffled words to the user and take his guess and then display the score.

In the data part the user has a set of options:

1. General and summarized weather data
2. Compare table
2. temperature in different units table
3. wind speed in different units.
4. sunset and sunrise.
5. friendly notifications about the weather state.
6. weather forecast where the user enter the number of days and it displays the forecast for the next entered number of days.

User Manual:

To engage with the application, users can choose between the entertainment or weather data options. Under the Entertainment category, users can opt for the encoding/decoding game by entering their username, decoding the shuffled word, and receiving a corresponding score. Alternatively, they can participate in a weather quiz, answering questions and obtaining a final score at the end.

1. .Click on the entertainment or weather data option
2. Entertainment: choose if you want the encoding/decoding game or a weather quiz and click on the desired button.
3. encoding/decoding game: enter your user name, then decode the shuffled word given and write it, and your score will be output

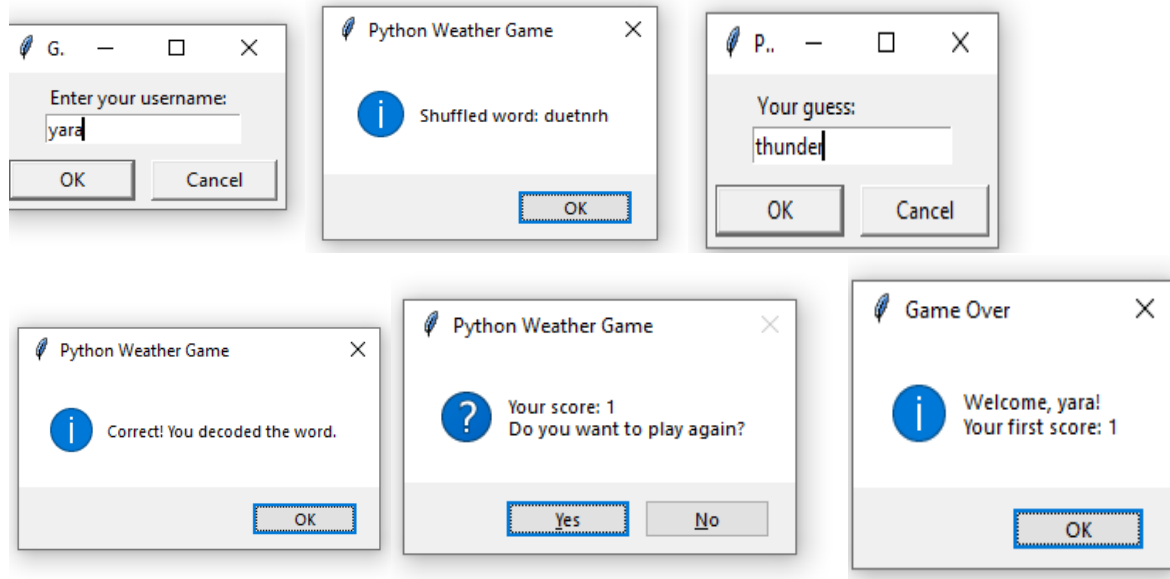


Figure (1).

4. weather quiz is: enter your username, then answer the questions by entering the number of the answer, your score is output at the end.

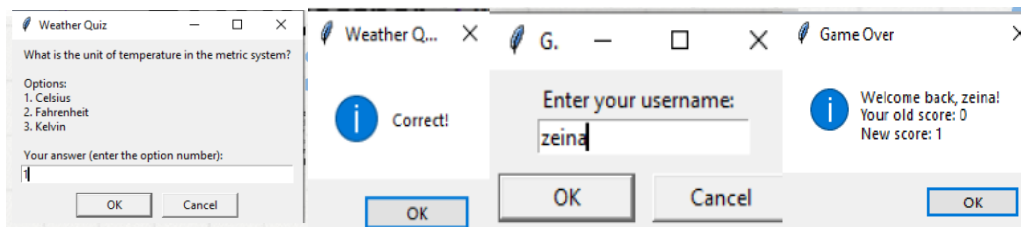


Figure (2).

5. If Weather data option is chosen, enter the city you're in the temperature, humidity, wind speed, and description of the weather is displayed and you will receive a notification containing a message depending on the temperature or description

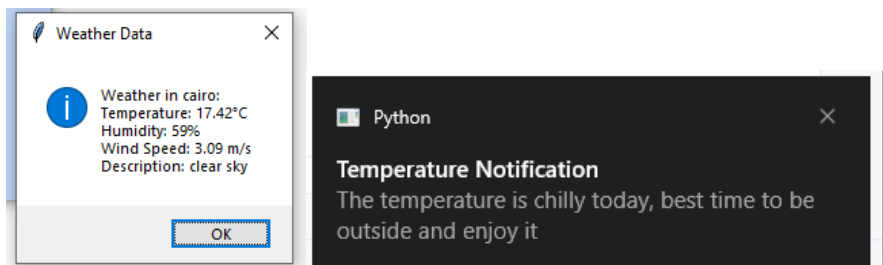


Figure (3).

6. To get more weather data of the city you entered, click “yes” when asked “Do you want more weather data?” 6 options will be displayed “Temperature table, Wind speed table, Sunset time, Sunrise, Compare, Weather forecast”
7. Temperature table: to display the temperature of the city entered in different units
 Wind speed Table: to display the wind speed of the city entered in different units
 Sunset time: to display the sunset time
 Sunrise: to display the sunrise time
 Compare: To compare the weather features between different cities
 Weather forecast: to display the temperature of the city entered over the desired number of days

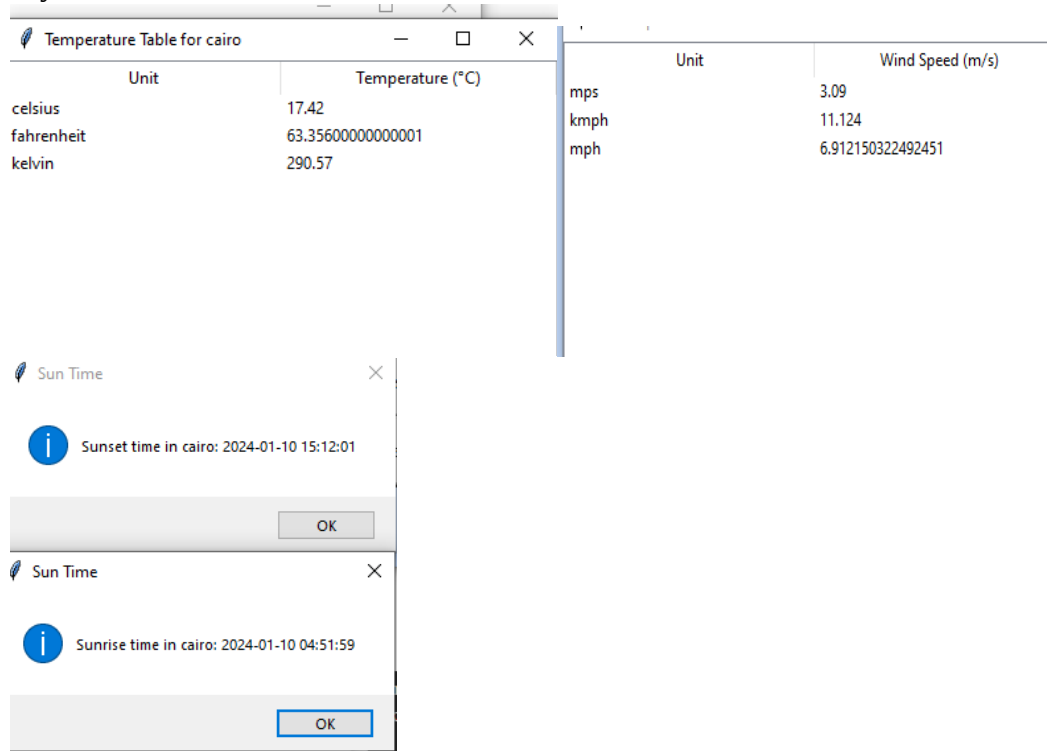


Figure (4).

8. Compare: enter the number of cities you want to compare the entered city at the beginning with, then enter the name of the cities, and the information will be displayed

City	Temperature (°C)	Humidity (%)	Wind Speed (m/s)	Description
cairo	17.42	59	3.09	clear sky
paris	-1.23	77	3.6	broken clouds
tokyo	3.59	58	10.29	broken clouds
london	2.2	73	3.6	broken clouds

Figure (5).

9. Weather forecast : enter the number of days you want the weather to be displayed in, then the forecast with the days will be output

Date	Temperature (°C)	Humidity (%)	Description
2024-01-10	17.42	59.0	clear sky
2024-01-11	15.41	57.25	few clouds
2024-01-12	14.98	53.5	few clouds
2024-01-13	15.9	37.38	few clouds
2024-01-14	14.72	61.75	clear sky

Figure (6).